

Quick Start for L^AT_EXing with IEEEtran.cls for IEEE Conferences

FirstName LASTNAME

Example University

firstname.lastname@example.edu

Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

I. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna.

Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus enim. Vestibulum pellentesque felis eu massa. TODO!

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on L^AT_EX (Section III). Finally, a conclusion is drawn and outlook on future work is made (Section IV).

II. RELATED WORK

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

III. L^AT_EX HINTS

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines. In case you want a new paragraph, just press enter twice! This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (\).

This is rarely used.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. A long motivation for that is provided at <http://loopSPACE.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3>.

Corresponding L^AT_EX code of sections/01-content.tex

```
91 One sentence per line.
92 This rule is important for the usage of version control systems.
93 A new line is generated with a blank line.
94 As you would do in Word:
95 New paragraphs are generated by pressing enter.
96 In LaTeX, this does not lead to a new paragraph as LaTeX joins
   subsequent lines.
97 In case you want a new paragraph, just press enter twice!
98 This leads to an empty line.
99 In word, there is the functionality to press shift and enter.
100 This leads to a hard line break.
101 The text starts at the beginning of a new line.
102 In LaTeX, you can do that by using two backslashes
   (\textbackslash\textbackslash).
103 \\
104 This is rarely used.
105
106 Please do \textit{not} use two backslashes for new paragraphs.
107 For instance, this sentence belongs to the same paragraph,
   whereas the last one started a new one.
108 A long motivation for that is provided at
   \url{http://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3}
```

B. Notes separated from the text

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding L^AT_EX code of sections/01-content.tex

```
116 \begin{mindflow}
117 This is a small note.
118 \end{mindflow}
```

C. Handling TODOs

Markierter Text.

Corresponding L^AT_EX code of sections/01-content.tex

```
124 \textmarker{Markierter Text.}
```

Bei `\textmarker` wird nur die Textfarbe geändert, da dies auch bei einigen Worten gut funktioniert.

Markierter Text.

Corresponding L^AT_EX code of sections/01-content.tex

```
130 \textcomment{Markierter Text.}{Kommentar dazu.}
```

Manuelle Markierung für Text, der seit der letzten Version geändert wurde.

Corresponding L^AT_EX code of sections/01-content.tex

```
134 \modified{Manuelle Markierung für Text, der seit der letzten
   Version geändert wurde.}
```

Das ist ein Text. **Geänderter Text.**

Corresponding L^AT_EX code of sections/01-content.tex

```
138 Das ist ein Text.
139 \change{FL1: Text angepasst}{Geänderter Text}.
```

Hier nur ein Kommentar.

Corresponding L^AT_EX code of sections/01-content.tex

```
143 Hier nur ein Kommentar\sidecomment{Kommentar}.
```

TODO!

Corresponding L^AT_EX code of sections/01-content.tex

```
147 \todo{Hier muss noch kräftig Text produziert werden}
```

D. Hyphenation

L^AT_EX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. It might be necessary to tweak the hyphenations nevertheless. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also write `\allowbreak` (result: application-specific), but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For instance, `application="specific` gets `application="specific`. This is enabled by an additional configuration of the babel package.

Corresponding L^AT_EX code of sections/01-content.tex

```
158 In case you write \enquote{application-specific}, then the word
   will only be hyphenated at the dash.
159 You can also write \verb!application\allowbreak!tion-specific1
   (result: applica\allowbreak!tion-specific), but this is
   much more effort.
160
161 You can now write words containing hyphens which are hyphenated
   at other places in the word.
162 For instance, \verb!application="specific1 gets
   application="specific.
163 This is enabled by an additional configuration of the babel
   package.
```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: $100 \frac{\text{km}}{\text{h}}$, or by using plain L^AT_EX (and math mode): $100 \frac{\text{km}}{\text{h}}$.

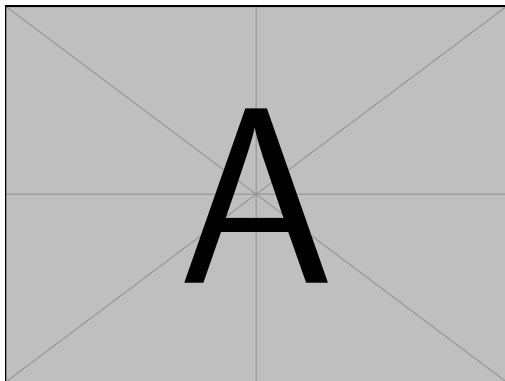


Figure 1. Example figure for cref demo

Heading1	Heading2
One	Two
Thee	Four

Figure 2. Example table for cref demo

Corresponding L^AT_EX code of sections/01-content.tex

```
169 Numbers can be written plain text (such as 100), by using the
    \href{https://ctan.org/pkg/siunitx}{siunitx} package as
    follows:
170 \SI{100}{\km\per\hour},
    or by using plain LATEX{} (and math mode):
171 $100 \frac{\mathit{km}}{h}$.
172
```

5% of 10 kg

Corresponding L^AT_EX code of sections/01-content.tex

```
176 \SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

Corresponding L^AT_EX code of sections/01-content.tex

```
180 Numbers are automatically grouped: \num{123456}.
```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding L^AT_EX code of sections/01-content.tex

```
186 Please use the \enquote{enquote command} to quote something.
187 Quoting with "quote" or `quote' also works.
```

G. Cleveref examples

Cleveref demonstration: Cref at beginning of sentence, cref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.



Figure 3. Simple Figure. Based on Scharrer [3].

Figure 2 shows a simple fact, although Figure 2 could also show something else.

Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding L^AT_EX code of sections/01-content.tex

```
218 \Cref{fig:ex:cref} shows a simple fact, although
    \cref{fig:ex:cref} could also show something else.
219
220 \Cref{tab:ex:cref} shows a simple fact, although
    \cref{tab:ex:cref} could also show something else.
221
222 \Cref{sec:ex:cref} shows a simple fact, although
    \cref{sec:ex:cref} could also show something else.
```

H. Figures

Figure 3 shows something interesting.

Corresponding L^AT_EX code of sections/01-content.tex

```
228 \Cref{fig:label} shows something interesting.
229
230 \begin{figure}
231   \centering
232   \includegraphics[width=.8\linewidth]{example-image-golden}
233   \caption[Simple Figure]{
234     Simple Figure.
235     Based on \cit{mwe}.
236   }
237   \label{fig:label}
238 \end{figure}
```

One can span a figure across multiple columns by using `\begin{figure*}`. See Figure 4 as an example.

Corresponding L^AT_EX code of sections/01-content.tex

```
245 \begin{figure*}
246   \centering
247   % note that \textwidth is used instead of \linewidth
248   % This ensures that the graphics width is 60% of the "page"
    (text block), and not just 60% of the current text column
249   % See https://tex.stackexchange.com/a/17085/9075 for details
250   \includegraphics[width=.6\textwidth]{example-image-16x9}
251   \caption{16x9 Figure}
252   \label{fig:16x9}
253 \end{figure*}
```

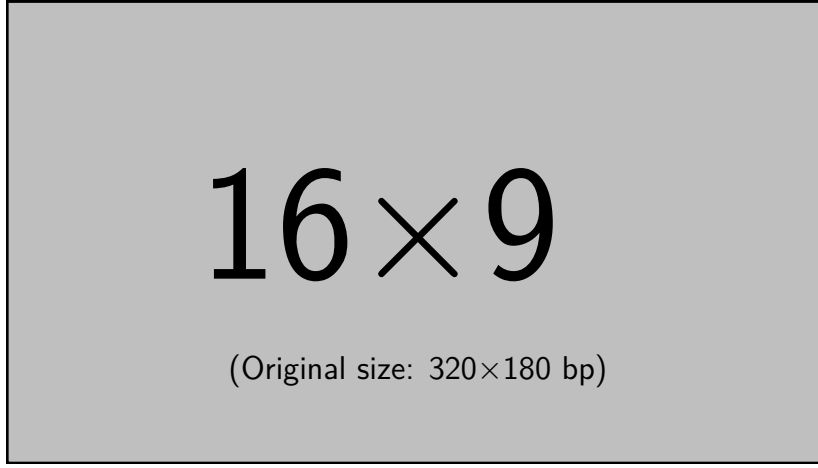


Figure 4. 16x9 Figure

I. Sub Figures

An example of two sub figures is shown in Figure 5.

Corresponding L^AT_EX code of sections/01-content.tex

```

262 \begin{figure*}[!b]
263 \centering
264 \subfloat[Case
      I]{\includegraphics[width=.4\linewidth]{example-image-a}%
265 \label{fig:first_case}}
266 \hfil
267 \subfloat[Case
      II]{\includegraphics[width=.4\linewidth]{example-image-b}%
268 \label{fig:second_case}}
269 \caption{Example figure with two sub figures.}
270 \label{fig:two_sub_figures}
271 \end{figure*}

```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate

the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, just leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 6.

Corresponding L^AT_EX code of sections/01-content.tex

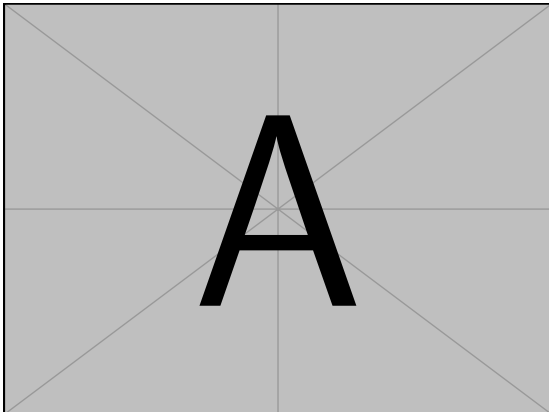
```

284 \begin{figure*}[!b]
285 \centering
286 \subfloat[]{\includegraphics[width=.4\linewidth]{example-image-a}%
287 \label{fig:first_case_ieee}}
288 \hfil
289 \subfloat[]{\includegraphics[width=.4\linewidth]{example-image-b}%
290 \label{fig:second_case_ieee}}
291 \caption{Example figure with two sub figures. IEEE style. (a)
      The first case. (b) The second case.}
292 \label{fig:two_sub_figures_ieee}
293 \end{figure*}

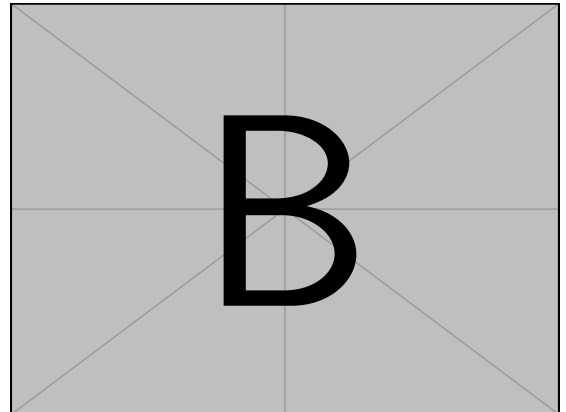
```

J. Tables

Note that IEEE does not support `\begin{table}`, one has to use `\begin{figure}`.



(a) Case I



(b) Case II

Figure 5. Example figure with two sub figures.

Corresponding L^AT_EX code of sections/01-content.tex

```

301 \begin{figure}
302   \caption{Simple Table}
303   \label{tab:simple}
304   \centering
305   \begin{tabular}{ll}
306     \toprule
307     Heading1 & Heading2 \\
308     \midrule
309     One      & Two      \\
310     Three    & Four      \\
311     \bottomrule
312   \end{tabular}
313 \end{figure}

```

Corresponding L^AT_EX code of sections/01-content.tex

```

317 % Source: https://tex.stackexchange.com/a/468994/9075
318 \begin{figure}
319   \caption{Table with diagonal line}
320   \label{tab:diag}
321   \begin{center}
322     \begin{tabular}{|l|c|c|}
323       \hline
324       \diagbox[width=10em]{Diag}{Column Head I}{Diag
325         Column}{Head II} & Second & Third \\
326       \hline
327       & foo & bar \\
328       \hline
329     \end{tabular}
330   \end{center}
331 \end{figure}

```

K. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```

1 <listing name="example">
2   <!-- comment -->
3   <content>not interesting</content>
4 </listing>

```

Listing 1. Example XML Listing

```

1 <listing name="example">
2   Floating
3 </listing>

```

Listing 2. Example XML listing – placed as floating figure

Corresponding L^AT_EX code of sections/01-content.tex

```

337 \Cref{lst:XML} shows source code written in XML.
338 \Cref{line:comment} contains a comment.
339
340 \begin{lstlisting}[
341   language=XML,
342   caption={Example XML Listing},
343   label={lst:XML}]
344   <listing name="example">
345     <!-- comment --> (* \label{line:comment} *)
346     <content>not interesting</content>
347   </listing>
348 \end{lstlisting}

```

One can also add `float` as parameter to have the listing floating. Listing 2 shows the floating listing.

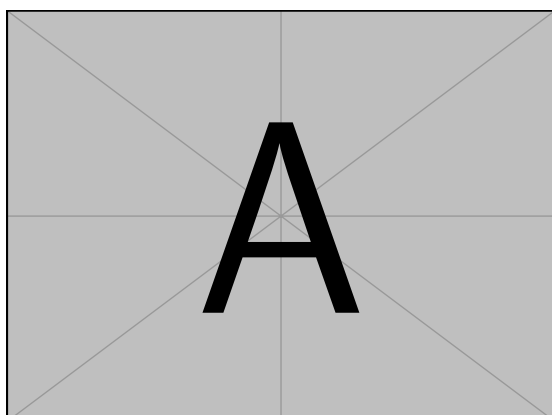
Corresponding L^AT_EX code of sections/01-content.tex

```

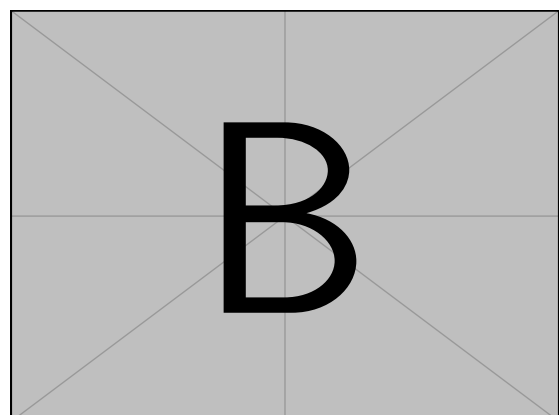
355 \begin{lstlisting}[
356   % one can adjust spacing here if required
357   % aboveskip=2.5\baselineskip,
358   % belowskip=-.8\baselineskip,
359   float,
360   language=XML,
361   caption={Example XML listing -- placed as floating figure},
362   label={lst:fXML}]
363   <listing name="example">
364     Floating
365   </listing>
366 \end{lstlisting}

```

One can also typeset JSON as shown in Listing 3.



(a)



(b)

Figure 6. Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

Figure 7. Simple Table

Heading1	Heading2
One	Two
Thee	Four

Figure 8. Table with diagonale line

Diag Column Head I	Diag Column Head II	Second	Third
		foo	bar

Corresponding L^AT_EX code of sections/01-content.tex

```

372 \begin{lstlisting}[
373     float,
374     language=json,
375     caption={Example JSON listing -- placed as floating figure},
376     label={lst:json}]
377 {
378     key: "value"
379 }
380 \end{lstlisting}
```

Java is also possible as shown in Listing 4.

Corresponding L^AT_EX code of sections/01-content.tex

```

386 \begin{lstlisting}[
387     caption={Example Java listing},
388     label=lst:java,
389     language=Java,
390     float]
391 public class Hello {
392     public static void main (String[] args) {
393         System.out.println("Hello World!");
394     }
395 }
396 \end{lstlisting}
```

L. Itemization

One can list items as follows:

- Item One
- Item Two

```

1 {
2     key: "value"
3 }
```

Listing 3. Example JSON listing – placed as floating figure

```

1 public class Hello {
2     public static void main (String[] args) {
3         System.out.println("Hello World!");
4     }
5 }
```

Listing 4. Example Java listing

Corresponding L^AT_EX code of sections/01-content.tex

```

404 \begin{itemize}
405     \item Item One
406     \item Item Two
407 \end{itemize}
```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding L^AT_EX code of sections/01-content.tex

```

413 \begin{compactitem}
414     \item Item One
415     \item Item Two
416 \end{compactitem}
```

One can enumerate items as follows:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of sections/01-content.tex

```

422 \begin{enumerate}
423     \item Item One
424     \item Item Two
425 \end{enumerate}
```

With the package paralist, one can create enumerations with lesser spacing:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of sections/01-content.tex

```

431 \begin{compactenum}
432     \item Item One
433     \item Item Two
434 \end{compactenum}
```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

1) All these items... 2) ...appear in one line 3) This is enabled by the paralist package.

Corresponding L^AT_EX code of sections/01-content.tex

```

440 \begin{inparaenum}
441     \item All these items...
442     \item ...appear in one line
443     \item This is enabled by the paralist package.
444 \end{inparaenum}
```

M. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding L^AT_EX code of sections/01-content.tex

```
450 The words \enquote{workflow} and \enquote{dwarflake} can be
    copied from the PDF and pasted to a text file.
```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1, 2, 3)$

Corresponding L^AT_EX code of sections/01-content.tex

```
454 The symbol for powerset is now correct: $\powerset$ and not a
    Weierstrass p ($\wp$).
455
456 $\powerset{\{1,2,3\}}$
```

Brackets work as designed: `<test>` One can also input backticks in verbatim text: ``test``.

Corresponding L^AT_EX code of sections/01-content.tex

```
460 Brackets work as designed:
461 <test>
462 One can also input backticks in verbatim text: \verb|`test`|.
```

IV. CONCLUSION AND OUTLOOK

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

ACKNOWLEDGMENT

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document [4].

REFERENCES

- [1] O. Kopp *et al.*, “Winery – A Modeling Tool for TOSCA-based Cloud Applications,” in *Proceedings of 11th International Conference on Service-Oriented Computing (ICSOC’13)*, ser. LNCS, vol. 8274. Springer Berlin Heidelberg, 2013, pp. 700–704.
- [2] T. Binz, G. Breiter, F. Leymann, and T. Spatzier, “Portable Cloud Services Using TOSCA,” *IEEE Internet Computing*, vol. 16, no. 03, pp. 80–85, May 2012.
- [3] M. Scharrer, *The mwe Package*, 2017. [Online]. Available: <http://texdoc.net/mwe>
- [4] B. Veytsman, “Latex class for the association for computing machinery – acknowledgement information,” Aug. 2021. [Online]. Available: <https://github.com/borisveytsman/acmart/blob/1704c8bf7ee92a1515f755f5118b6a22bb1f8e/samples/samples.dtx#L709>

All links were last followed on October 5, 2020.